

Which warehouse serves each customer — T1 running example

This document explains the example used in the T1 slides. It contains all the data and all the results shown on the slides, so that anyone can follow and check them. The data are invented for this course (2026); they do not describe a real company.

1. The situation

A company distributes pallets from two warehouses, W1 and W2, to four customers, C1 to C4. Each week it must decide which warehouse serves each customer. **Each customer is served entirely from one warehouse:** its order is not split between the two.

Table 1. Warehouse capacity

Warehouse	Capacity (pallets/week)
W1	80
W2	70

Table 2. Forecast demand

Customer	Forecast demand (pallets/week)
C1	40
C2	30
C3	35
C4	25
Total	130

Table 3. Shipping cost (€/pallet)

	C1	C2	C3	C4
W1	2	3	2	6
W2	6	4	7	2

Table 4. Delivery time (days)

	C1	C2	C3	C4
W1	1	1	1	3
W2	2	4	3	1

For every customer, the nearest warehouse is both the cheapest and the fastest: W1 for C1, C2 and C3; W2 for C4.

2. The problem in mathematical form

What is decided. For each warehouse w and customer c , whether w serves c :

$$x_{wc} = \begin{cases} 1 & \text{if warehouse } w \text{ serves customer } c \\ 0 & \text{otherwise} \end{cases}$$

Data. $W = \{W1, W2\}$ is the set of warehouses and $C = \{C1, C2, C3, C4\}$ the set of customers; d_c is the forecast demand of customer c (pallets/week); K_w is the capacity of warehouse w (pallets/week); k_{wc} is the shipping cost from w to c (€/pallet).

What is sought. The lowest total shipping cost:

$$\min \sum_{w \in W} \sum_{c \in C} k_{wc} d_c x_{wc}$$

What limits it.

- Each customer is served by exactly one warehouse:

$$\sum_{w \in W} x_{wc} = 1 \quad \forall c \in C$$

- No warehouse ships more than its capacity:

$$\sum_{c \in C} d_c x_{wc} \leq K_w \quad \forall w \in W$$

- Each variable is 0 or 1:

$$x_{wc} \in \{0, 1\} \quad \forall w \in W, \forall c \in C$$

This is an assignment problem with capacities. It is related to the transport problem of T3-T; the difference is that in the transport problem a customer's demand may be split between several warehouses.

3. Deciding from experience versus deciding with the model

Rule of thumb: each customer from its nearest warehouse. W1 would have to ship $40 + 30 + 35 = 105$ pallets, but it can only ship 80. The plan is not feasible.

Diverting by hand. A common fix: take the customers one by one, in order C1, C2, C3, C4; each goes to its nearest warehouse, and if that warehouse has no room left, to the other one. C1 and C2 fit in W1 (70 pallets); C3 does not fit ($70 + 35 = 105 > 80$) and is sent to W2; C4 goes to W2. This is plan M.

The model. Solving the problem of §2 gives plan A: C3 stays in W1 and C2 is the one sent to W2.

Table 5. Plan from experience and plan from the model

Plan	C1	C2	C3	C4	Load W1	Load W2	Cost (€/week)
M · diverted by hand	W1	W1	W2	W2	70	60	465
A · model	W1	W2	W1	W2	75	55	320

The model saves 145 €/week (31 % of the cost of plan M). The reason: sending C3 to W2 costs 5 € more per pallet (7 instead of 2) on 35 pallets, that is 175 €; sending C2 to W2 costs only 1 € more per pallet (4 instead of 3) on 30 pallets, that is 30 €. The rule of thumb diverts whichever customer happens to come last, not the one that is cheapest to divert.

4. How many plans are there?

Each of the 4 customers can be served by either of the 2 warehouses, so there are $2 \times 2 \times 2 \times 2 = 2^4 = 16$ possible plans. Only 5 of them respect the capacities.

Table 6. All 16 plans, ordered by cost (feasible plans first)

Plan	C1	C2	C3	C4	Load W1	Load W2	Feasible	Cost (€/week)	Average delivery time (days)
A	W1	W2	W1	W2	75	55	yes	320	1.69
B	W2	W1	W1	W2	65	65	yes	450	1.31
M	W1	W1	W2	W2	70	60	yes	465	1.54
D	W2	W2	W1	W1	60	70	yes	580	2.38
E	W1	W2	W2	W1	65	65	yes	595	2.62
—	W1	W1	W1	W2	105	25	no (W1)	290	1.00
—	W1	W1	W1	W1	130	0	no (W1)	390	1.38
—	W1	W2	W1	W1	100	30	no (W1)	420	2.08
—	W2	W2	W1	W2	35	95	no (W2)	480	2.00
—	W1	W2	W2	W2	40	90	no (W2)	495	2.23
—	W2	W1	W1	W1	90	40	no (W1)	550	1.69
—	W1	W1	W2	W1	95	35	no (W1)	565	1.92
—	W2	W1	W2	W2	30	100	no (W2)	625	1.85
—	W2	W2	W2	W2	0	130	no (W2)	655	2.54
—	W2	W1	W2	W1	55	75	no (W2)	725	2.23
—	W2	W2	W2	W1	25	105	no (W2)	755	2.92

Sixteen plans can be checked by hand. In general, with m warehouses and n customers there are m^n plans. With 10 warehouses and 200 customers there are 10^{200} plans, a number with 201 digits: they cannot be listed, not even by a computer. This is why the solution methods of T2 are needed.

5. Two objectives: cost and delivery time

So far a plan has been judged only by its cost. The company also wants its customers served fast, and the cheapest plan is not the fastest.

The second objective. A pallet of customer c sent from warehouse w takes t_{wc} days to arrive (Table 4). The average delivery time of a plan is the number of days that a pallet takes on average:

$$T(x) = \frac{\sum_{w \in W} \sum_{c \in C} t_{wc} d_c x_{wc}}{\sum_{c \in C} d_c}$$

In the numerator, $t_{wc} d_c x_{wc}$ counts the pallet-days of customer c : its d_c pallets, each taking t_{wc} days, if w serves it ($x_{wc} = 1$). The denominator is the total number of pallets, 130. For plan A:

$$T = \frac{40 \cdot 1 + 30 \cdot 4 + 35 \cdot 1 + 25 \cdot 1}{130} = \frac{220}{130} = 1.69 \text{ days}$$

The problem with two objectives. The model of §2 gets a second objective; the decisions and the constraints do not change:

$$\begin{aligned} \min \quad & Z_1(x) = \sum_{w \in W} \sum_{c \in C} k_{wc} d_c x_{wc} && (\text{cost, €/week}) \\ \min \quad & Z_2(x) = \frac{1}{\sum_c d_c} \sum_{w \in W} \sum_{c \in C} t_{wc} d_c x_{wc} && (\text{average delivery time, days}) \\ \text{s.t.} \quad & \sum_{w \in W} x_{wc} = 1 && \forall c \in C \\ & \sum_{c \in C} d_c x_{wc} \leq K_w && \forall w \in W \\ & x_{wc} \in \{0, 1\} && \forall w \in W, \forall c \in C \end{aligned}$$

There is usually no plan that is best on both objectives. Solving the problem has two steps: discard the plans that no one should choose, and choose among the others.

Table 7. The five feasible plans on both criteria

Plan	Cost (€/week)	Average delivery time (days)	Beaten by
A	320	1.69	none: no feasible plan is cheaper
B	450	1.31	none: no feasible plan is faster
M	465	1.54	B, which is cheaper ($450 < 465$) and faster ($1.31 < 1.54$)
D	580	2.38	A, B and M, each cheaper and faster
E	595	2.62	A, B, M and D, each cheaper and faster

Step 1: Pareto-optimal plans. A plan beats another if it is no worse on both criteria and better on at least one. A feasible plan is Pareto-optimal if no other feasible plan beats it (T3-T). A plan that is beaten should never be chosen, because another plan is at least as cheap and at least as fast.

- A is Pareto-optimal because no feasible plan is cheaper, and B because no feasible plan is faster.
- Plan M, the one diverted by hand, is beaten by B. D and E are beaten by A, among others.
- Plans that do not fit in the warehouses do not count: the plan that serves each customer from its nearest warehouse costs 290 €/week and takes 1.00 days, better than all of them, but it is not feasible.

The choice is therefore between A, the cheaper, and B, the faster. Going from A to B costs $450 - 320 = 130$ €/week more and shortens the average delivery by $1.69 - 1.31 = 0.385$ days (exactly $50/130$). The model cannot say whether that is worth it: it depends on what a faster delivery is worth to the company.

Step 2: choosing with a price for time. One way to choose is to give time a price. Let V be what the company would pay, in euros per week, to make the average delivery one day shorter. Its unit is €/week per day: euros per week, like the cost, for each day of average delivery time that is saved. $V = 100$, for example, means that shortening the average delivery by one day is worth 100 €/week to the company, and by half a day, 50 €/week. The value comes from the company: fewer complaints, sales that are not lost, smaller stocks at the customers. With V the two objectives become one, in €/week, with the same constraints:

$$\min \quad Z(x) = Z_1(x) + V \cdot Z_2(x)$$

This is the weighted sum method: V is the weight given to the delivery time. The model is of the same kind as that of §2 and is solved in the same way. With $V = 100$, plan A has $Z = 320 + 100 \times 1.69 = 489$ €/week and plan B $Z = 450 + 100 \times 1.31 = 581$ €/week, so A is chosen.

Where the choice changes. B saves 0.385 days for 130 €/week more: each day saved costs $130/0.385 = 338$ €/week. If a day is worth less than that to the company ($V < 338$ €/week per day), A is chosen; if it is worth more ($V > 338$), B is chosen. At $V = 338$ both have the same total. M, D and E are never chosen, whatever V : each is beaten by a plan that is cheaper and faster, which always has a lower total.

In a figure. If the plans are drawn with the average delivery time on the horizontal axis and the cost on the vertical one, all the points with the same total Z lie on the line $\text{cost} = Z - V \times \text{time}$, of slope $-V$: along it, one day more of delivery time is compensated by V euros less of cost. The best plan is the feasible plan on the lowest such line. A larger V gives a steeper line, and above 338 the lowest line goes through B instead of A. The notebook of this example draws this line and lets V be changed (§8).

The weighted sum is one of several ways of choosing between objectives; another is to minimise the cost with a limit on the delivery time (T3-T).

6. Uncertainty: the forecast can be wrong

The plan is built on forecast demand. Table 8 shows the last 12 weeks of orders of customer C3. The forecast used is the average of the last four weeks (a 4-week moving average).

Table 8. Orders of C3 and forecast error (pallets/week)

Week	1	2	3	4	5	6	7	8	9	10	11	12
Orders	33	36	34	38	31	35	37	32	36	34	35	35

Week	1	2	3	4	5	6	7	8	9	10	11	12
Forecast	—	—	—	—	35.25	34.75	34.50	35.25	33.75	35.00	34.75	34.25
Error (orders – forecast)	—	—	—	—	–4.25	0.25	2.50	–3.25	2.25	–1.00	0.25	0.75

Over weeks 5–12:

- Mean absolute error (MAE): 1.81 pallets/week.
- Mean absolute percentage error (MAPE): 5.4 %.
- Bias (average error): –0.31 pallets/week, close to zero, so the forecast neither over- nor under-estimates systematically.

The forecast for week 13 is $(36 + 34 + 35 + 35)/4 = 35$ pallets, the value used in Table 2.

What happens if the error is large. In week 13, C3 orders 46 pallets, 11 more than forecast (31 % more). In plan A, W1 serves C1 and C3: $40 + 46 = 86$ pallets, but its capacity is 80. Six pallets of C3 cannot be served from W1. Plan A had only 5 pallets of spare capacity in W1.

What a safety margin costs. If each warehouse keeps 10 % of its capacity in reserve (W1 plans with 72 pallets, W2 with 63), plans A and B are no longer feasible and the cheapest plan becomes M, at 465 €/week: the margin costs 145 €/week.

Rolling horizon. In practice the plan is not fixed once: each week new orders arrive, the forecast is updated and the plan is recomputed. This is called planning with a rolling horizon (T3-P).

§7 keeps a plan for several weeks while the orders change, and measures what that costs.

7. Plans under uncertain demand

§6 looks at one week: the forecast of C3 fails in week 13, and a reserve of capacity costs money. This section keeps a plan for several weeks while the orders of two customers change. It checks every feasible plan week by week, measures the forecast error of two customers, and computes what each plan costs over a period when a pallet that is not served has a cost.

The data. The orders of C1 are now also data of the example (Table 9). They are invented for this course (2026), like the rest of the example. C2 and C4 order the same every week, by contract: 30 and 25 pallets. Only C1 and C3 have orders that change. In week 13, C1 orders 40 pallets, the same as its forecast, and C3 orders 46, as in §6.

Table 9. Orders of C1 and C3 (pallets/week)

Week	1	2	3	4	5	6	7	8	9	10	11	12	13
C1	38	41	43	39	44	37	42	45	40	36	43	41	40
C3	33	36	34	38	31	35	37	32	36	34	35	35	46

The assumptions.

1. The plan, which warehouse serves each customer, is kept for a period of N weeks, with no changes.
2. Each customer is served from one warehouse. The pallets above the capacity of a warehouse in a week are not served: they are lost (they are not served in another week), and each one costs p € to the company.
3. Each week is independent of the others: no stock and no pending orders pass from one week to the next.
4. The cost of a plan in the period is its weekly cost of Table 6 (computed with the forecast demand of Table 2) times N , plus p for each pallet not served. The change in the shipping cost when a customer orders more or less than forecast is not counted.

The plans week by week. In week t , with D_{ct} the order of customer c , the *load* of warehouse w is the pallets of the customers it serves, its *margin* is its capacity minus its load, and the pallets *not served* are those above its capacity:

$$L_{wt} = \sum_{c \in C} D_{ct} x_{wc}, \quad \text{margin}_{wt} = K_w - L_{wt}, \quad \text{not served}_{wt} = \max(0, L_{wt} - K_w)$$

A negative margin is a load above capacity. Table 10 sums up the five feasible plans of Table 6 over weeks 1 to 12, and gives the pallets not served in week 13 (C1 40, C2 30, C3 46, C4 25).

Table 10. The feasible plans week by week (pallets/week)

Plan	Weeks 1–12 with a load above capacity	Smallest margin W1 / W2, weeks 1–12	Not served in week 13
A	none	1 / 15	6 (W1)
B	none	12 / 0	0
M	none	5 / 7	1 (W2)
D	2, 3, 5, 7, 8, 11, 12 (W2)	17 / –5	0
E	none	10 / 2	6 (W2)

The forecast and its error, for C1 and C3. The forecast of each customer is the 4-week moving average of §6, and its error is orders – forecast. Besides MAE, MAPE and bias, the *largest positive error* is the largest amount by which the orders exceeded the forecast in weeks 5 to 12:

$$e_{\max}^+ = \max_{t=5, \dots, 12} e_t$$

Table 11 gives the errors and the four measures of both customers.

Table 11. Forecast error of C1 and C3, weeks 5–12

Customer	Errors, weeks 5–12 (pallets)	MAE (pallets/week)	MAPE	Bias (pallets/week)	Largest positive error (pallets)
C1	3.75, –4.75, 1.25, 4.50, –2.00, –5.00,	2.94	7.3 %	0.00	4.50 (week 8)

Customer	Errors, weeks 5–12 (pallets)	MAE (pallets/week)	MAPE	Bias (pallets/week)	Largest positive error (pallets)
	2.25, 0.00				
C3	–4.25, 0.25, 2.50, –3.25, 2.25, –1.00, 0.25, 0.75	1.81	5.4 %	–0.31	2.50 (week 7)

The forecast for week 13 is 40.00 pallets for C1 and 35.00 for C3, the demand of Table 2.

When two customers whose orders change share a warehouse, their errors add up in it. In plan A, W1 serves C1 and C3; the sum of their errors in weeks 5 to 12 is –0.50, –4.50, 3.75, 1.25, 0.25, –6.00, 2.50 and 0.75 pallets, and the largest is 3.75 (week 7). With the forecast demand, W1 has 5 pallets of spare capacity in plan A (§6). In the other four plans, C1 and C3 are served from different warehouses.

The cost of a plan over a period. Over a period of N weeks, the cost of a plan is

$$\text{total cost} = N \cdot c + p \cdot U$$

where:

- c is the weekly cost of the plan (€/week, Table 6);
- N is the number of weeks of the period;
- p is the cost of a pallet that is not served (€/pallet);
- U is the number of pallets not served in the period.

In the weeks of the period without a surprise, every customer orders its forecast demand and every feasible plan serves all its pallets. In s weeks with a surprise, C1 and C3 order more or less than forecast (by default, 40 and 46 pallets, as in week 13). Then $U = s \cdot u$, where u is the number of pallets the plan does not serve in one week with a surprise.

Table 12 is an example with $N = 12$ weeks, $p = 100$ €/pallet and one week with a surprise, in which C1 orders 40 and C3 orders 46 pallets.

Table 12. Cost of each plan over 12 weeks ($p = 100$ €/pallet, one week with a surprise)

Plan	Weekly cost c (€/week)	Not served in the week with a surprise, u (pallets)	Total cost $N \cdot c + p \cdot U$ (€)
A	320	6	4,440
B	450	0	5,400
M	465	1	5,680
D	580	0	6,960
E	595	6	7,740

The ranking of the plans depends on p , N , s and the orders in a week with a surprise.

Doing these calculations. Everything in this section can be computed by hand with the formulas above and the data of Tables 2, 6 and 9. Section 7 of the [notebook](#) does them for any plan, customer and values of N , p and s , and the two scripts of §8 print Tables 9 to 12.

8. Reproducing the numbers

Two Python scripts in this folder reproduce every table of this document. Running them is optional: it is not part of what is assessed in the course. How to install and run each one is explained at the top of the file.

Script	How it finds the best plan	What must be installed
warehouse_allocation.py	It checks all 16 plans and keeps the cheapest feasible one (brute force)	Nothing beyond Python
warehouse_allocation_solver.py	It writes the model of §2 and lets a solver find the best plan, without listing the plans	The libraries PuLP and HiGHS (pip install pulp highspy)

Both print the same results. With 16 plans, checking them all is quick. With 10^{200} plans (§4) it is impossible. A solver does not need to list the plans, so it can often solve problems of that size: how it does so is the subject of the optimisation methods of T2.

The notebook [notebooks/warehouse_allocation.ipynb](#) goes through §3–§7 step by step, with forms to try your own plans, the value of a faster delivery, other orders of C3, other safety margins and, in its section 7, plans kept for several weeks under uncertain demand. It opens in Google Colab without installing anything:  [Open in Colab](#). Its code is in the package [warehouses](#) of this folder ([src/warehouses/](#)).